

State-Dependent Inhibition of Nav1.8 Sodium Channels by VX-150 and VX-548[§]

Patric Vaelli, Akie Fujita,  Sooyeon Jo, Han-Xiong Bear Zhang, Tomás Osorno, Xiao Ma, and Bruce P. Bean

Department of Neurobiology (P.V., A.F., S.J., H.-X.B.Z., T.O., B.P.B.) and Laboratory of Systems Pharmacology and Harvard Program in Therapeutics (X.M.), Harvard Medical School, Boston, Massachusetts

Received May 4, 2024; accepted September 16, 2024

ABSTRACT

Nav1.8 sodium channels (Nav1.8) are an attractive therapeutic target for pain because they are prominent in primary pain-sensing neurons with little expression in most other kinds of neurons. Recently, two Nav1.8-targeted compounds, VX-150 and VX-548, have shown efficacy in clinical trials for reducing pain. We examined the characteristics of Nav1.8 inhibition by these compounds. The active metabolite form of VX-150 (VX-150m) inhibited human Nav1.8 channels with an IC_{50} of 15 nM. VX-548 (suzetrigine) was even more potent (IC_{50} 0.27 nM). Both VX-150m and VX-548 had the unusual property of “reverse use-dependence,” whereby inhibition could be relieved by repetitive depolarizations, a property seen before with another Nav1.8 inhibitor, A-887826. The relief of VX-548 inhibition by large depolarizations occurred with a time constant of ~40 milliseconds that was not concentration-dependent. Reinhibition at negative voltages occurred with a rate that was nearly proportional to drug concentration, consistent with the idea that relief of inhibition reflects dissociation of drug from the channel and reinhibition

reflects rebinding. The relief of inhibition by depolarization suggests a remarkably strong and unusual state-dependence for both VX-150m and VX-548, with very weak binding to channels with fully activated voltage sensors despite very tight binding to channels with voltage sensors in the resting state.

SIGNIFICANCE STATEMENT

The Nav1.8 sodium channel (Nav1.8) is a current target for new drugs for pain. This work describes the potency, selectivity, and state-dependent characteristics of inhibition of Nav1.8 channels by VX-150 and VX-548, compounds that have recently shown efficacy for relief of pain in clinical trials but whose mechanism of interaction with channels has not been described. The results show that the compounds share an unusual property whereby inhibition is relieved by depolarization, demonstrating a state-dependence different from most sodium channel inhibitors.

Introduction

Nav1.8 sodium channels (Nav1.8) are neuronal sodium channels with a distinctive pattern of expression, being strongly expressed in C-fiber pain-sensing neurons (nociceptors) and only a few other types of neurons (Han et al., 2016; Bennett et al., 2019; Goodwin and McMahon, 2021; Jayakar et al., 2021). The strong expression of Nav1.8 channels in nociceptors, along with their prominent role in supporting action potential firing of nociceptors, makes them an attractive pharmacological target, and a number of Nav1.8 inhibitors have been developed (Payne et al., 2015; Brown et al., 2019; Urru et al., 2020; reviewed by Han et al., 2016; Yekkirala et al., 2017; Obeng et al., 2021). Recently, two new Nav1.8

inhibitors, VX-150 (Hijma et al., 2021) and VX-548 (Jones et al., 2023), showed efficacy and safety in human clinical trials for reducing acute pain.

The mechanism by which VX-150m and VX-548 interact with Nav1.8 channels has not yet been reported. Many small-molecule sodium channel inhibitors interact with channels in a strongly state-dependent manner. Most commonly, inhibitors bind more tightly to channels when channels are in a depolarized state so that channel inhibition is enhanced during repetitive depolarization, the phenomenon of “use-dependence” typical of many local anesthetics, antiarrhythmics, and anti-seizure drugs (Hille, 1977), including many compounds that inhibit Nav1.8 channels, including vinpocetine (Zhou et al., 2003), carbamazepine (Cardenas et al., 2006), tetracaine (Browne et al., 2009a), and ambroxol (Leffler et al., 2010). Strikingly, two previously studied Nav1.8 inhibitors, A-803467 (Jarvis et al., 2007) and A-887826 (Zhang et al., 2010), showed a phenomenon of “reverse use-dependence,” where inhibition was partially relieved by repetitive depolarizations (Browne et al., 2009a,b; Jo et al., 2023). The effect was only modest with A-803467 (Browne et al., 2009a,b; Jo et al., 2023) but dramatic with A-887826 (Jo et al., 2023).

This work was supported by National Institutes of Health National Institute of Neurological Disorders and Stroke [Grants R01-NS036855, R01-NS110860, and R35-NS127216] and the Defense Advanced Research Projects Agency [Award HR0011-19-2-0022].

No author has an actual or perceived conflict of interest with the contents of this article.

dx.doi.org/10.1124/molpharm.124.000944.

[§] This article has supplemental material available at molpharm.aspetjournals.org.

ABBREVIATIONS: cryo-EM, cryogenic electron microscopy; I_{max} , maximum current; I_{min} , minimum current; k, slope factor; K_d , dissociation constant; Nav1.8, Nav1.8 sodium channel; V_h , midpoint.

Here we have explored state-dependent interactions with Nav1.8 channels of the more potent inhibitors VX-150 and VX-548. We find that both compounds show dramatic relief of inhibition by depolarization, suggesting greatly reduced affinity for binding to channels with voltage sensors in a depolarized position.

Materials and Methods

Pharmacology. The structure of the VX-150 prodrug form (denoted as VX-150p) was obtained from the patent literature (Ryan, 2020), and the structure of the active metabolite (denoted as VX-150m) was inferred from the structure of the prodrug. Both forms were synthesized as described in Supplemental Material 1 ("Synthesis of VX-150 prodrug and metabolite"). The structure of VX-548 (suzetrigine) was inferred from the patent literature (Karkare, 2022) and synthesized as described in the Supplemental Material 2 ("Synthesis of VX-548"). The final step in the synthesis was to separate two enantiomers on a chiral column. The active enantiomer was determined by electrophysiological recordings and corresponded to the second peak coming from the column. The active enantiomer is designated as "VX-548." VX-150m and VX-548 were prepared as 10-mM or 1-mM stock solution in DMSO (Sigma-Aldrich) and stored as frozen aliquots at -20°C . A-887826 was obtained from Sigma-Aldrich, prepared as a stock solution of 10 mM in DMSO, and stored as frozen aliquots at -20°C .

Human Nav1.8 Cell Line. A cell line stably expressing human Nav1.8 α - and β 3 subunits in Chinese hamster ovary (CHO-K1) cells was purchased from B'SYS GmbH and grown in Ham's F-12 medium (Corning) containing 10% FBS, penicillin/streptomycin (Sigma), and 3.5 $\mu\text{g}/\text{ml}$ puromycin (Sigma) and 350 $\mu\text{g}/\text{ml}$ hygromycin (Sigma) under 5% CO_2 at 37°C .

Cell Preparation for Electrophysiology. Prior to recording, cell cultures were washed twice with divalent-free phosphate-buffered saline (Corning) and incubated in Detachin enzyme solution (Genlantis) for 3 minutes at 37°C . The cell enzyme mixture was diluted 1:10 in serum-free Ham's F-12 media and centrifuged at 65G for 2 minutes. The cell pellet was resuspended in 1 ml of "EC000" extracellular solution (in mM, 145 NaCl, 4 KCl, 2 CaCl_2 , 1 MgCl_2 , 10 HEPES, 10 glucose, pH 7.4 with NaOH) supplemented with 50 mM glucose and 1 mg/ml Pluronic-F68 (Sigma). The cells were pelleted again and resuspended in supplemented EC000 solution. Cell density and viability were measured with a Countess 3 cell counter (Invitrogen). Cell density was $3\text{--}8 \times 10^6$ cells/ml (most commonly $4\text{--}5 \times 10^6$ cells/ml) with viability of 97%–99%.

Electrophysiology Using the QPatch Compact. Voltage clamp recordings were made using the QPatch Compact automated patch clamp system (Sophion). Recordings were made using multihole plates (QPlate 8X) in which whole-cell recordings of up to 10 cells are obtained simultaneously in each of the eight wells of the instrument. Recordings were made at ambient temperature, which was somewhat warmer inside the instrument than room temperature; measurement of plate temperature using an infrared thermometer gave a reading of 27°C immediately after removal from the instrument. Recordings were obtained using a CsF-based internal solution (140 mM CsF, 10 mM NaCl, 10 mM HEPES, 1 mM EGTA, pH 7.3 adjusted with CsOH, osmolarity adjusted to 320 mOsm with sucrose) and a base external solution of EC000. The liquid junction potential between these solutions is +9.3 mV (external relative to internal), calculated using the stationary Nernst-Planck equation (Marino et al., 2014) using LjPcalc (RRID:SCR_025044; <https://sw Harden.com/LjPcalc/>), and not corrected for. All external solutions contained 0.1% Pluronic F-68 (Sigma), a surfactant poloxamer previously found to enable more sensitive assays of lipophilic compounds when using the low-volume wells in automated patch clamp instruments (Houtmann et al., 2017). In an initial series of experiments performed without Pluronic F-68, the apparent potencies of both VX-150m and VX-548 were substantially reduced (for VX-150m, IC_{50} of 45 nM without Pluronic

and 15 nM with Pluronic; for VX-548, IC_{50} of 3 nM without Pluronic and 0.27 nM with Pluronic).

Sophion Analyzer v.6.7.54 software was used for controlling the operation of the QPatch Compact and for data acquisition. The whole-cell protocol for forming seals and obtaining whole-cell recordings was based on the "CHO - Sophion standard" protocol with whole-cell detection level modified to 6 pF from default value of 4 pF. Sodium currents from a single well were typically 3–15 nA.

Sodium currents were evoked using depolarizations from a standard holding potential of -100 mV using a variety of voltage protocols designed to test state-dependence, as shown in the figures. Currents were digitized at 50 kHz after filtering with an eighth-order Bessel filter with a corner frequency of 16.66 kHz. After digitization and acquisition, currents and command waveforms were exported from the Sophion Analyzer software (v.6.7.54) and imported into Igor Pro (version 6.7, WaveMetrics) for analysis.

Data Analysis and Statistics. Voltage-dependent sodium current was measured after raw currents were corrected for linear capacitative and leak currents, which were determined using 10 mV hyperpolarizations delivered from the resting potential and then appropriately scaled and subtracted assuming linearity.

To quantify the voltage-dependence with which inhibition was relieved by 200-millisecond depolarizing pulses in the protocols shown in Figs. 3, 5, and 8, current during the second test pulse (after a 30-millisecond return to -100 mV to allow recovery from fast inactivation) was plotted as a function of the voltage of the 200-millisecond prepulse and fit with a Boltzmann function ($I_{\min} + (I_{\max} - I_{\min}) * (1 / (1 + \exp(-(V_m - V_h)/k)))$) to data from minimum to maximum, where $I_{\min}$ is the minimum current, $I_{\max}$ is the maximum current, V_m is the voltage of the prepulse, V_h is the midpoint, and k is the slope factor.

All experiments were exploratory rather than hypothesis testing, and statistical results of mean $\pm$ S.D. are descriptive and presented to enable evaluation of reproducibility (Michel et al., 2020). Statistical tests on the differences in voltage-dependence and kinetics of relief among VX-548, VX-150m, and A-887826 were made using the nonparametric two-tailed Mann-Whitney test (Marx et al., 2016).

Results

Inhibition by VX-150m. VX-150 is a prodrug given to patients in a form containing a phosphate group that is hydrolyzed by endogenous phosphatases to produce a more active metabolite (Fig. 1A). Initial experiments showed that the prodrug form (VX-150p) inhibited sodium currents in the human Nav1.8-expressing stable cell line with a half-blocking concentration of about 300 nM and that the metabolite (VX-150m) was much more potent. Figure 1, B and C shows the inhibition of sodium current in the Nav1.8-expressing cell line by VX-150m. VX-150 m at 30 nM produced more than 50% inhibition. VX-150m at 1 μM produced saturating effects, with 10 μM having no further effect.

There was a small inward current remaining with 1 μM and 10 μM VX-150m. This current had faster activation and inactivation kinetics than the main current in the cells and was inhibited by 1 μM tetrodotoxin, consistent with representing a small endogenous sodium current from a TTX-sensitive channel type that was revealed when the dominant Nav1.8 current was inhibited by VX-150m. Collected data gave a dose-response curve consistent with complete inhibition of Nav1.8 current by VX-150m with a dissociation constant (K_d) of 15 nM, taking account of the remaining TTX-sensitive current that comprised on average 6% of the total current.

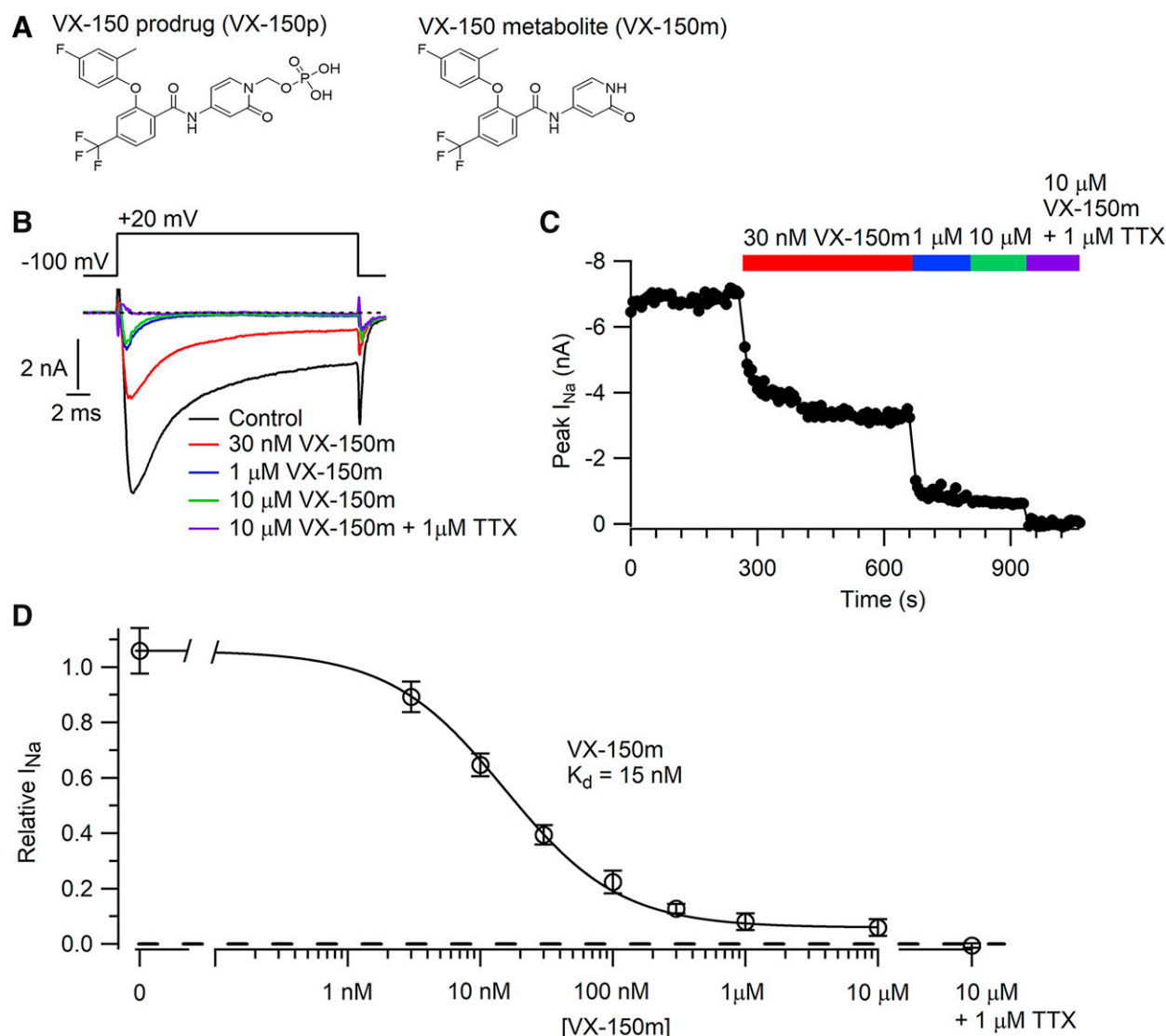


Fig. 1. Concentration-dependence of VX-150m inhibition of sodium current in a cell line expressing human Nav1.8 channels. (A) Structures of the prodrug (VX-150p) and inferred active metabolite (VX-150m). (B) Inhibition of current evoked by a 20-millisecond depolarization from -100 mV to $+20$ mV by application of VX-150m at 30 nM, 1 μ M, and 10 μ M, followed by 1 μ M tetrodotoxin in the continued presence of 10 μ M VX-150m. (C) Time course of inhibition in this experiment. (D) Collected results, mean $\pm$ S.D. $N = 11$ for 0 (relative current measured after 7 minutes of exposure to control drug-free solution with 0.01% DMSO), $n = 4$ for 3 nM VX-150m, $n = 6$ for 10 nM, $n = 11$ for 30 nM, $n = 7$ for 100 nM, $n = 8$ for 300 nM, $n = 23$ for 1 μ M, $n = 8$ for 10 μ M, $n = 8$ for 10 μ M VX-150m with 1 μ M TTX. Solid line: Fit to $I_{min} + (I_{max} - I_{min}) / (1 + [Drug]/K_d)$, with I_{max} set to mean relative current measured after 7 minutes of exposure to control drug-free solution with 0.01% DMSO (1.06 ± 0.08 , $n = 11$), I_{min} set to mean current measured in 10 μ M VX-150m (0.06 ± 0.03 , $n = 8$) and $K_d = 15$ nM. Dashed line indicates zero.

Reverse Use-Dependence of VX-150m Inhibition. We explored whether VX-150m produces use-dependent inhibition like that seen with compounds like carbamazepine (Cardenas et al., 2006) or tetracaine (Browne et al., 2009a) or “reverse use-dependence” (partial relief of inhibition by repeated depolarizations) like that produced by A-803467 (Browne et al., 2009a,b) and more dramatically by A-887826 (Jo et al., 2023). With currents evoked by 20-millisecond depolarizations to $+30$ mV delivered at 20 Hz, in control solution there was a gradual reduction in current amplitude during 4 seconds of stimulation at 20 Hz to 0.70 ± 0.03 of initial current (mean $\pm$ S.D., $n = 6$), presumably reflecting gradual accumulation of a fraction of channels into a slow inactivated state (Fig. 2). After inhibition by VX-150m, there was clear reverse use-dependence. With

30 nM VX-150m, current during the 4-second stimulation period increased from 0.30 ± 0.05 (compared with initial resting control) at the beginning to 0.44 ± 0.05 at the end of the 4-second train (mean $\pm$ S.D., $n = 6$). With 100 nM VX-150m, current increased from 0.11 ± 0.02 to 0.22 ± 0.04 during the train (mean $\pm$ S.D., $n = 6$).

Relief of VX-150m Inhibition by Long Depolarizing Steps. The reverse use-dependence with VX-150m implies that inhibition can be relieved by depolarization. To explore this phenomenon, we tested the ability of longer steps to a range of voltages to relieve VX-150m inhibition. We used a protocol with 200-millisecond steps to a range of voltages between -100 mV to $+120$ mV, followed immediately by a test step to $+10$ mV and then by a 30-millisecond return to -100 mV before a second test step to $+10$ mV. In separate experiments

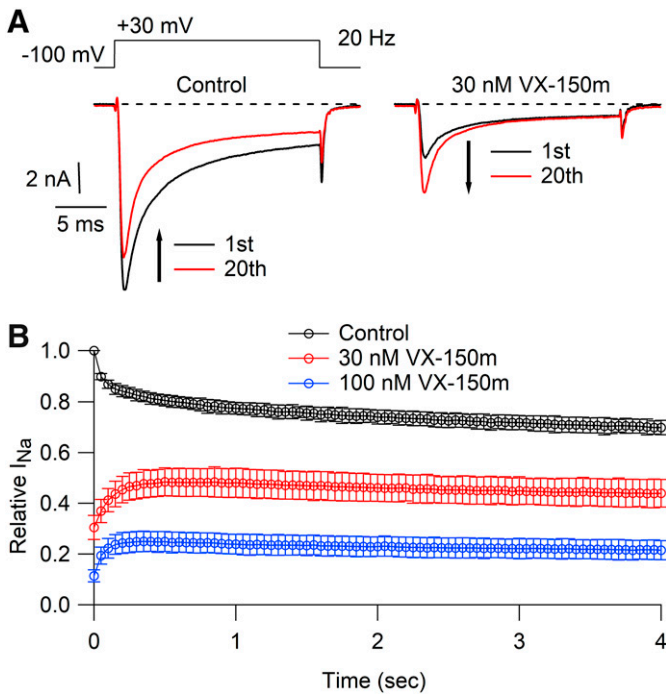


Fig. 2. Use-dependent relief of VX-150m inhibition of Nav1.8 channels. (A) First (black) and 20th (red) currents evoked by a series of 20-millisecond steps to +30 mV delivered at 20 Hz in control (left) and 30 nM VX-150m (right). (B) Collected results with this protocol. Current was normalized to the first current in control solution before application of drug. Points plot mean $\pm$ S.D. ($n = 6$) for each condition.

in control conditions using short depolarizations to produce only fast inactivation, we determined that the 30-millisecond return to -100 mV allows complete recovery from fast inactivation. With the 200-millisecond prepulse protocol in control conditions, current during the second test pulse showed reduction by about 35% as the 200-millisecond prepulse became more depolarized (Fig. 3A), reflecting the entry of channels into slow inactivated states. However, after inhibition by 30 nM VX-150m, the current during the second test pulse increased as the prepulse became more depolarized (Fig. 3B), as if inhibition was relieved.

Figure 3C shows the current during the test pulse immediately after the prepulse as a function of the voltage during the prepulse. In control, test pulse current declines monotonically to about 15% of current from -100 mV, reflecting inactivation of the channels. In the presence of VX-150m, about 75% of channels were inhibited at the resting potential of -100 mV, and the remaining current during the test pulse declined over the range from -70 to -10 mV with a similar voltage-dependence as in control, consistent with inactivation of the fraction of channels without drug bound. However, with prepulses from -10 mV to $+120$ mV, the first test pulse current actually increased and reached a value similar to the asymptotic value in control. This apparent relief of inhibition by strong depolarizations was very striking when assayed by the second test pulse after the 30-millisecond period at -100 mV to allow recovery from fast inactivation (Fig. 3D), with current during the second test pulse increasing dramatically over the prepulse voltage range of -20 mV to $+120$ mV, reaching a value of about 95% of the control current after strong depolarizations. Thus, VX-150m inhibition can be almost completely removed by

200-millisecond depolarizations to voltages positive to about $+60$ mV.

Inhibition by VX-548. We next examined inhibition by VX-548, a compound that has been reported to be more potent as a Nav1.8 inhibitor and that has shown efficacy in human clinical trials for acute pain (Jones et al., 2023). We inferred the likely structure of VX-548 from the patent literature (Karkare, 2022). The synthesized molecule is stereoisomeric, and the enantiomers were separated on a chiral column. We tested the activity of both enantiomers as Nav1.8 inhibitors and found that one enantiomer produced potent Nav1.8 inhibition at subnanomolar concentrations and the other produced no inhibition up to a concentration of at least $10 \mu\text{M}$.

The active enantiomer produced exceptionally potent inhibition of Nav1.8 channels. In these experiments, having discovered with VX-150m that the cell line expresses a small amount of non-Nav1.8 TTX-sensitive current, we added 200 nM TTX to all solutions to inhibit this component of current. The active enantiomer applied at 1 nM inhibited current to about 25% of control, and 30 nM completely inhibited the current (Fig. 4). The dose-response of Nav1.8 inhibition by the active enantiomer could be fit reasonably well, assuming a 1:1 interaction with a dissociation constant of 0.27 nM. As this is somewhat more potent than the IC_{50} of 0.7 nM reported for recombinant Nav1.8 inhibition by VX-548 in the publication of the clinical results (Jones et al., 2023), we assume that the structure of the compound that we synthesized corresponds to the clinically used VX-548, and we therefore assume that it is correct to refer to this compound as VX-548. The difference in apparent potency from the previously published value can likely be explained by small differences in preparing or applying solutions containing such low concentrations of compound. In our studies, we initially found an IC_{50} of about 3 nM, and it was only after discovering that using solutions containing 1 mg/ml Pluronic F-68 resulted in greatly enhanced potency that dramatic effects of subnanomolar concentrations were evident.

In the Nav1.8 cell line that we used, inactivation was incomplete, with a substantial component of current remaining at the end of the 20-millisecond depolarization to $+20$ mV that we used as a standard test pulse. This is consistent with studies that have noted a relatively large component of noninactivating persistent current from human Nav1.8 compared with rat Nav1.8 channels (Han et al., 2015). Both VX-548 and VX-150m inhibited the component of noninactivating current with similar efficacy as inhibition of peak current.

Relief of VX-548 Inhibition by Depolarization. We next tested whether VX-548 inhibition was affected by depolarization, using the same prepulse protocol as in Fig. 3. Like VX-150m inhibition, inhibition by VX-548 was very effectively relieved by 200-millisecond depolarizations, with strong depolarizations to $+120$ mV producing nearly complete reversal of inhibition (Fig. 5).

Kinetics of Relief and Reinhibition by VX-548. There are two general possibilities for the ability of VX-548-inhibited channels to open after strong depolarizations. One is that the compound stays bound to the channel and that compound-bound channels can open and conduct ions with sufficiently strong depolarizations, with compound remaining bound to the channel. The other is that movement of the voltage-sensor regions of the channel produce a conformational

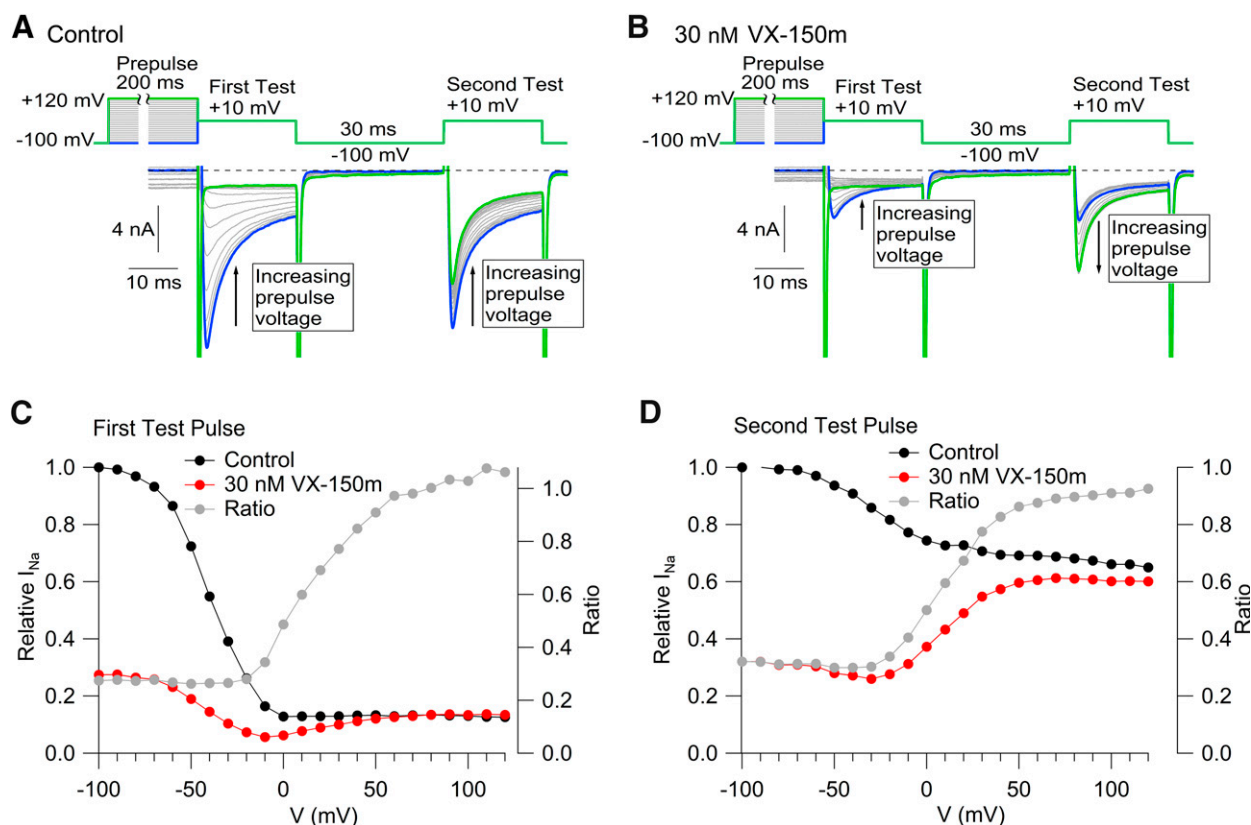


Fig. 3. Relief of VX-150m inhibition by single 200-millisecond depolarizations. (A) Voltage protocol for assaying relief of inhibition. A 200-millisecond prepulse to varying voltages is followed by a first test pulse to +10 mV, which assays channel availability immediately after the prepulse. A second test pulse to assay availability is given after returning to -100 mV for 30 milliseconds to allow channels to recover from fast inactivation. In control, channel availability tested by the second test pulse decreases slightly as a result of channels entering slow inactivation during strong prepulses. (B) Currents measured with the same protocol after inhibition by 30 nM VX-150m. VX-150m decreases channel availability to about 25% when assayed by a test step from -100 mV; in VX-150m, channel availability assayed by the second test step increases when prepulses are depolarized beyond about -20 mV. (C) Voltage-dependence of channel availability measured by the first test pulse in control (black) and in 30 nM VX-150m (red), with current in VX-150m normalized to maximal current (from -100 mV) in control. Gray symbols show ratio of currents in VX-150m and control. (D) Same for channel availability assayed by the second test pulse. Collected data for five repetitions of this experiment are shown in Fig. 8B.

change that results in weakened affinity for the compound and that the compound dissociates from the channel. We did further experiments examining the kinetics and concentration-dependence with which VX-548 inhibition is relieved at strong depolarizations and the kinetics with which reinhibition occurs at negative membrane potentials after relief. These experiments support the idea that VX-548 dissociates from channels at strong depolarizations. Figure 6 shows a protocol designed to test the kinetics with which VX-548 is relieved. A series of depolarizations are given to +120 mV, a voltage where there is almost full relief of inhibition by 1–3 nM VX-548 with 200-millisecond long depolarizations. The depolarizations to +120 mV are given with increasing duration, starting with a short (3-millisecond) step. After each step, the membrane is returned to -100 mV for 30 milliseconds to allow recovery from inactivation, and a test step to 0 mV then assays the fraction of sodium channels that are drug-free and able to open. Plotting the current in each test pulse as a function of the total cumulative time at +120 mV showed that VX-548 inhibition was relieved at +120 mV with a time course that could be fit very well by a single exponential function. The time constant of the relief was very similar with concentrations of VX-548 of 3 nM, 10 nM, and 30 nM (Fig. 6B), with values of 32 to 43 milliseconds.

The experiment whose results are shown in Fig. 7 examined the rate at which inhibition was reestablished when the membrane voltage was returned to -100 mV after producing maximal relief by a 300-millisecond step to +120 mV, again using an efficient single sweep protocol. Interestingly, the rate of reinhibition was very strongly dependent on the concentration of VX-548. In all cases, the time course of reinhibition could be fit well by a single exponential function. In this example, the time constant of reinhibition decreased from 16 seconds with 1 nM VX-548 to 0.9 seconds with 10 nM VX-548. The strong dependence of the rate of reinhibition on the concentration of VX-548 is consistent with the idea that VX-548 dissociates from channels during strong depolarizations and then rebinds when channels return to their resting state at -100 mV.

Comparison of Voltage-Dependent Relief of Inhibition and Reverse Use-Dependence by VX-548, VX-150m, and A-887826. Figure 8 shows collected results for the voltage-dependence with which VX-548 inhibition is relieved by depolarization and compares with collected results with VX-150m and A-887826. In all cases, inhibition could be relieved nearly completely by 200-millisecond depolarizations to +90 or +100 mV, but the voltage-dependence of relief was different for the three inhibitors. VX-150m inhibition required

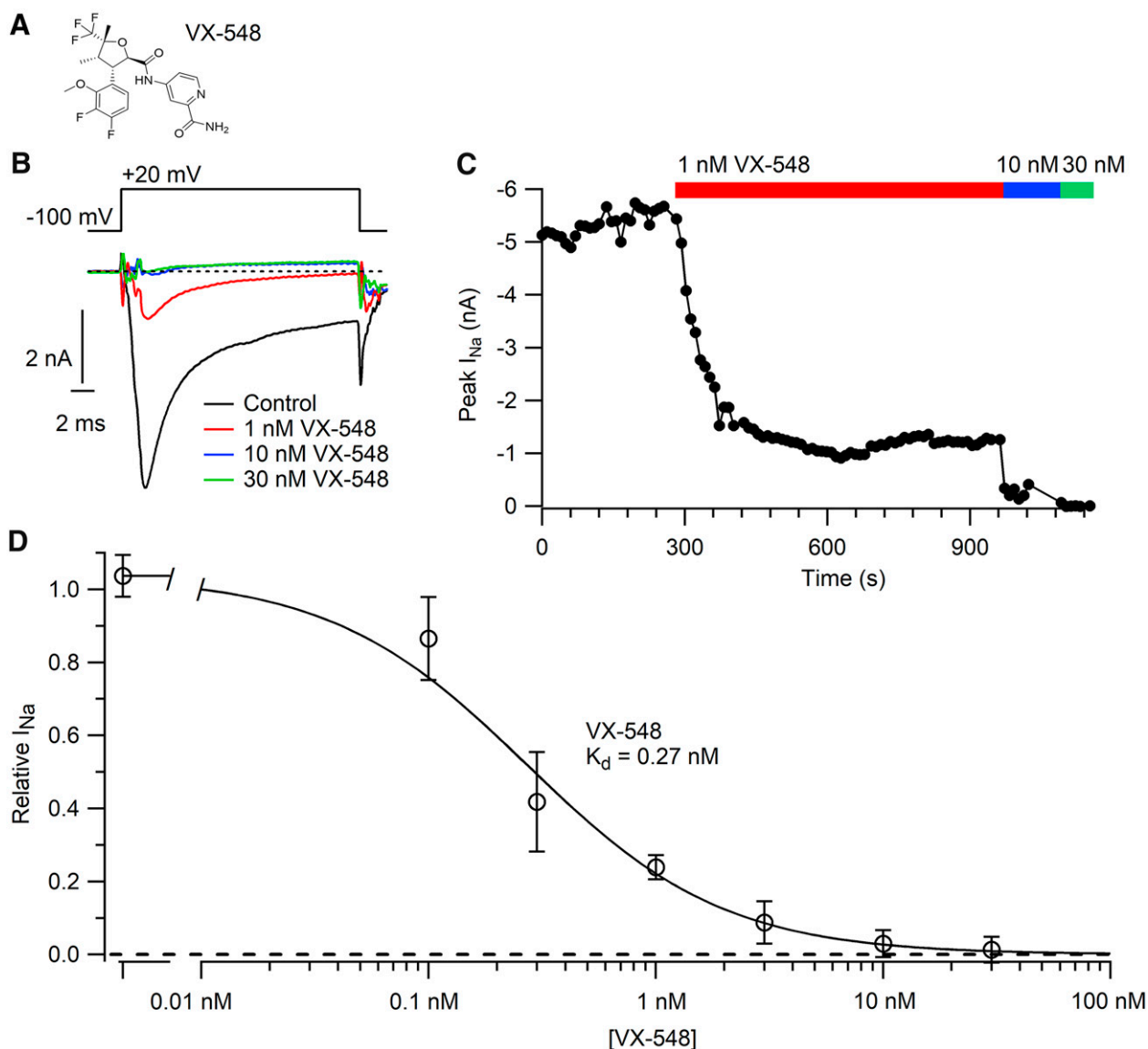


Fig. 4. Inhibition of human Nav1.8 channels by VX-548. (A) Structure of VX-548. (B) Inhibition of current evoked by a 20-millisecond depolarization from -100 mV to +20 mV by application of VX-548 at 1 nM, 10 nM, and 30 nM. All solutions contained 200 nM TTX to inhibit endogenous TTX-sensitive current. (C) Time course of inhibition in this experiment. (D) Collected results, mean $\pm$ S.D. $N = 5$ for 0 (relative current measured after 10 minutes of exposure to control drug-free solution with 0.000033% DMSO, corresponding to the DMSO concentration in the 0.3 nM VX-548 solution), $n = 5$ for 0.1 nM, $n = 7$ for 0.3 nM, $n = 8$ for 1 nM, $n = 7$ for 3 nM, $n = 8$ for 10 nM, $n = 17$ for 30 nM. Solid line: Fit to $I_{max}/(1 + [Drug]/K_d)$, with I_{max} set to mean relative current measured after 10 minutes of exposure to control drug-free solution with DMSO (1.04 ± 0.06 , $n = 5$), and $K_d = 0.27$ nM. Dashed line indicates zero.

the least strong depolarizations for relief, with a midpoint of $+3.7 \pm 4.3$ mV for the effect (mean $\pm$ S.D., $n = 5$). Relief of inhibition by A-887826 required much stronger depolarizations, with a midpoint of $+34.3 \pm 5.4$ mV (mean $\pm$ S.D., $n = 4$), and relief of inhibition by VX-548 required even stronger depolarizations, with a midpoint of $+56.7 \pm 4.6$ mV (mean $\pm$ S.D., $n = 5$). These differences were statistically significant ($P = 0.008$ for VX-548 vs. VX-150m; $P = 0.016$ for VX-548 vs. A-887826; $P = 0.016$ for VX-150m vs. A-887826, two-tailed Mann-Whitney test).

The three inhibitors also had very different kinetics of relief of inhibition (Fig. 9A). As shown in Fig. 6 for VX-548, relief during depolarization developed with a time course that could be fit well by a single exponential, with little dependence on the concentration of the inhibitor. Figure 9A shows the time-constant of relief with VX-548, VX-150m, and A-887826.

Relief of inhibition was much faster with VX-150m (time-constant of 2.9 ± 0.4 milliseconds with 30 nM VX-150m; mean $\pm$ S.D., $n = 6$) than with A-887826 (time-constant of 12.2 ± 2.9 milliseconds with 300 nM A-887826, $n = 4$; $P = 0.0095$, two-tailed Mann-Whitney test) and was slowest with VX-548 (time-constant of 39.2 ± 2.3 milliseconds with 30 nM VX-548, $n = 8$; $P = 0.0007$ relative to 30 nM VX-150m, $P = 0.004$ relative to 300 nM A-887826). It is interesting that the speed of relief, which is most simply interpreted as the rate of unbinding from the depolarized state of the channel, does not correlate with the IC_{50} for inhibition at negative holding potentials, which reflects binding affinity to the resting state of the channel. VX-548 has the highest binding affinity to the resting state, with a K_d of 0.27 nM, and also the slowest unbinding from the depolarized state, which would be consistent with the strongest binding to both states. Tighter binding of VX-548 to the depolarized state compared

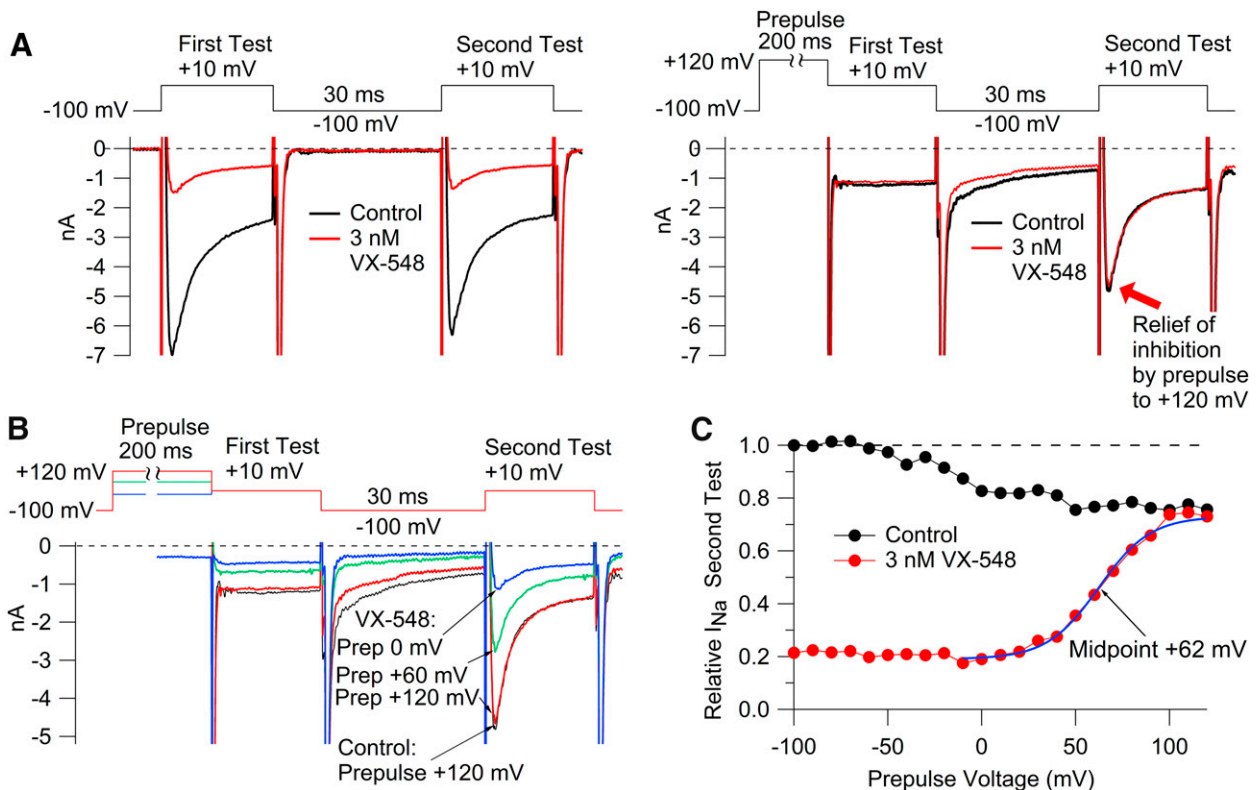


Fig. 5. Relief of VX-548 inhibition by depolarization. (A) Currents evoked by test pulses from -100 to $+10$ mV before and after inhibition by 3 nM VX-548 (left) and in same recording, currents evoked by the test pulses after a 200 -millisecond prepulse to $+120$ mV (right), illustrating nearly complete relief of inhibition by the prepulse. (B) Relief of inhibition in 3 nM VX-548 produced by 200 -millisecond prepulses to 0 mV (blue), $+60$ mV (green), and $+120$ mV (red), compared with current in control after a 200 -millisecond prepulse to $+120$ mV (black, partly overlaid by red trace in VX-548). (C) Voltage-dependence of relief of VX-548 inhibition by 200 -millisecond prepulses. Solid line: Fit to data in 3 nM VX-548 between -10 and $+120$ mV with Boltzmann function with a midpoint $+62$ mV and a slope factor of 13.7 mV. Collected data for five repetitions of this experiment are shown in Fig. 8A.

with VX-150m or A-887826 is consistent with the requirement for large depolarizations for relief of inhibition, as shown in Fig. 8. However, although VX-150m has a higher affinity for binding to the resting state (K_d of 15 nM) than A-887827 ($K_d \sim 100$ nM with the same conditions), VX-150m apparently has weaker binding than A-887826 for binding to the depolarized state, as reflected by both a much faster rate of relief and by requiring much smaller depolarizations (Fig. 8, B and C).

Figure 9B shows the kinetics of reinhibition after channels return to the resting state at -100 mV after relief by a depolarization. For all three, the kinetics of reinhibition could be fit well by a single exponential function, as shown for VX-548 in Fig. 7, and for all the inhibitors, the rate of reinhibition was roughly proportional to the concentration of the inhibitor. However, the rate of reinhibition by A-887826 stood out by being very slow compared with VX-548 or VX-150m. For example, the time-constant for reinhibition by 100 nM A-887826 (46 ± 15 seconds, mean $\pm$ S.D., $n = 4$) was more than 500 times that for 100 nM VX-150m (88 ± 16 milliseconds, $n = 6$; $P = 0.0095$, two-tailed Mann-Whitney test.) Interestingly, the data for the rate of reinhibition by VX-548 and VX-150m were roughly continuous as a function of drug concentration (Fig. 9B) and could be reasonably fit by a line of slope 1 , giving an on-rate constant of 1.06×10^8 seconds $^{-1}$ M $^{-1}$. In striking contrast, the much slower kinetics of A-887826 reinhibition were reasonably well fit with an on-rate constant of 1.2×10^5 seconds $^{-1}$ M $^{-1}$.

The relief of inhibition by strong depolarizations seen with VX-548 suggests that there could be reverse use-dependence with repeated shorter and smaller depolarizations, as for A-887826 (Jo et al., 2023) and VX-150m (Fig. 2). However, the extent of such reverse use-dependence depends on the voltage-dependence and time-dependence of relief and also the kinetics of reinhibition between depolarizing pulses. Figure 10 compares reverse use-dependence for the three compounds using the same voltage protocol. Examining reverse use-dependence with repeated 20 -millisecond depolarizations to $+30$ mV, using concentrations of each compound designed to produce resting inhibition to roughly similar levels of 5% – 10% of control current, reverse use-dependence was most dramatic with A-887826, with an increase in current from 0.05 ± 0.02 (compared with initial resting control) before the 4 -second train of depolarizations to 0.40 ± 0.06 at the end of the train ($n = 4$, mean $\pm$ S.D.). The increase with A-887826 occurred gradually over the entire 4 -second period, as if each depolarization produced a small amount of recovery that was not completely lost by reinhibition after the return to -100 mV before the next depolarization. Use-dependent relief with VX-548 was substantially smaller, with an increase in current from 0.12 ± 0.04 (compared with initial resting control) before the 4 -second train of depolarizations to 0.25 ± 0.06 at the end of the train ($n = 5$). As for A-887826, the kinetics of use-dependent relief were fairly slow, with relief accumulating gradually over the 4 -second period of stimulation. Use-dependent relief from VX-150m inhibition had very different kinetics,

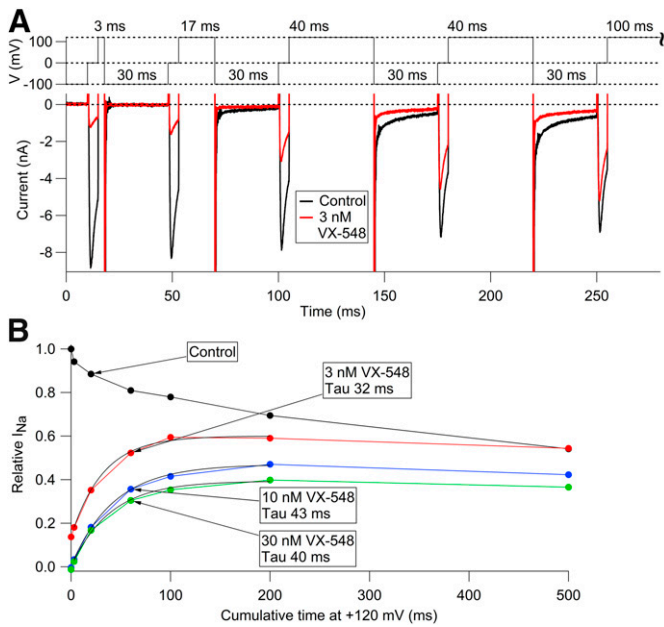


Fig. 6. Time-course of relief of VX-548 inhibition in different concentrations of VX-548. (A) Fraction of drug-free channels was assayed by short (5-millisecond) test steps to 0 mV, each after a 30-millisecond period at -100 mV to allow recovery from fast inactivation. A succession of depolarizations to +120 mV produced progressive slow inactivation in control (black trace) and progressive relief of inhibition in VX-548 (red trace). The protocol continued with increasing durations of the steps to +120 mV. (B) Channel availability during the test steps to 0 mV as a function of cumulative time at +120 mV in control (black), 3 nM VX-548 (red), 10 nM VX-548 (blue), and 30 nM VX-548 (green). Data points are fitted with single exponential functions with the indicated time constants. Collected data from eight repetitions of this experiment are shown in Fig. 9A.

with a rapid initial increase from 0.11 ± 0.02 in the first pulse of the train to a maximum of 0.25 ± 0.04 in the eighth pulse, followed by a gradual decrease to 0.22 ± 0.04 at the end of the 4-second period ($n = 6$). The decrease that follows the initial increase is similar in overall kinetics to the decrease that is seen in the absence of inhibitor, which presumably reflects gradual accumulation of channels in a slow inactivated state, with a small amount of entry into slow inactivation during each depolarization and incomplete recovery during the return to -100 mV before the next depolarization.

The differences in voltage-dependences of relief of inhibition and the different time-dependence of relief and reinhibition shown in Figs. 8 and 9 help explain why A-887826 gives the most dramatic reverse use-dependence: although both the voltage-dependence and time-dependence of relief of inhibition by A-887826 are intermediate between VX-150m and VX-548, the far slower reinhibition between pulses means that relief of inhibition continues to build up continuously during trains of depolarization.

Discussion

These results show that the new Nav1.8 inhibitors VX-150 and VX-548 both have the unusual property that inhibition is dramatically reduced by depolarization. All of the results are consistent with the simple idea that when channels change their state when the membrane is depolarized, the depolarized state of the channel has such a low affinity for

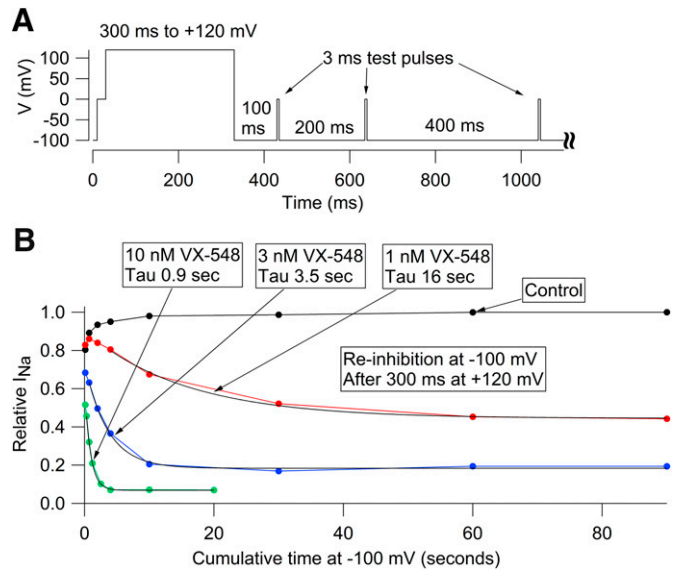


Fig. 7. Time-course of VX-548 reinhibition after relief of inhibition. (A) Voltage protocol for examining the time course of reinhibition in a single sweep. Relief of inhibition is produced by a 300-millisecond step to +120 mV, and channel availability is assayed by a series of test pulses to 0 mV delivered at various times after the long prepulse. The test pulses are short to minimize any entry into slow inactivation. (B) Time-course of reinhibition in different concentrations of VX-548 applied successively in the same recording. Peak current during each test pulse is normalized to maximal current availability in control, determined by a test pulse to 0 mV preceding the 300-millisecond prepulse in control. Data points are fitted with single exponential functions with the indicated time constants. Collected data from eight repetitions of this experiment are shown in Fig. 9B.

the drug that the drug dissociates from the channel. The idea that relief of inhibition represents dissociation of drug instead of slow activation of channels with drug still bound is supported by the fact that reinhibition upon repolarization occurs at a rate roughly proportional to the drug concentration, as expected if drug must rebind. In contrast, if relief of inhibition represented slowed opening of channels with drug still bound—as occurs with some gating-modifying peptide toxins (McDonough et al., 1997; Swartz and MacKinnon, 1997; Swartz, 2007)—there is no reason that reinhibition would depend on drug concentration.

The results imply that the affinity of drug binding to the channel undergoes an enormous change when channels convert from the resting state to the depolarized state. The change is especially dramatic for VX-548, where the affinity of the resting state is remarkably high, with a dissociation constant of 0.27 nM. The affinity for the depolarized state must be more than 100 times lower because inhibition by 30 nM VX-548 can be reversed by about 70% by a 500-millisecond depolarizing prepulse (Fig. 6).

A recent study (Gilchrist et al., 2024) showed that depolarizing steps produced partial relief of inhibition by another Nav1.8 inhibitor, LTGO-33, a compound that is closely related structurally to VX-150. That study used mutagenesis to suggest that the compound binds at a site formed by an extracellular cleft formed by S1, S3, and S4 helices along with the S3-S4 linker region of the voltage sensor region of domain II (Gilchrist et al., 2024). Because depolarization induces an outward movement of the S4 helix, it is easy to see that the conformation of such a binding pocket could be dramatically

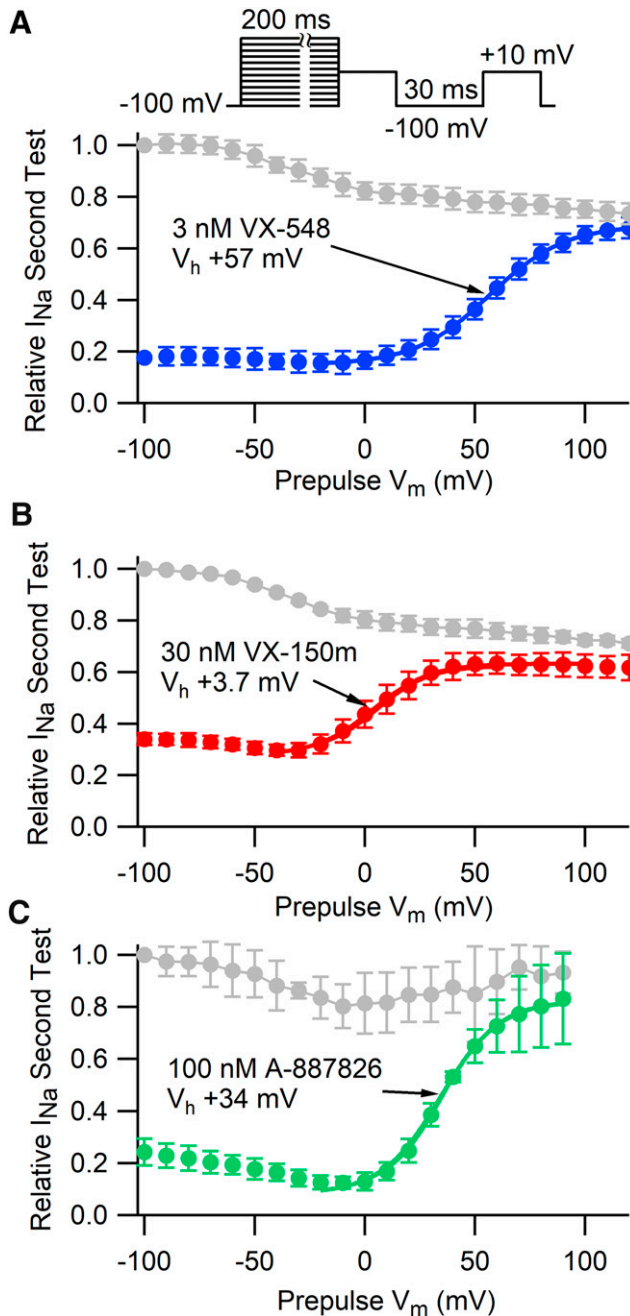


Fig. 8. Comparison of voltage-dependence of relief of inhibition by VX-548, VX-150m, and A-887826. (A) Collected results for voltage-dependence of relief of inhibition by depolarization with 3 nM VX-548 determined using the protocol shown in Fig. 5. Gray symbols: current during second test pulse in control normalized to the value with a prepulse to -100 mV, showing voltage-dependence of slow inactivation (mean $\pm$ S.D., $n = 5$). Blue symbols: current during the second test pulse after application of 3 nM VX-548, with current in VX-548 normalized to maximal current (from -100 mV) in control. Solid line: Boltzmann function $(I_{min} + (I_{max} - I_{min}) * (1 / (1 + \exp(-(V_m - V_h) / k))))$, where I_{min} is the minimum current, I_{max} is the maximum current, V_m is the voltage of the prepulse, V_h is the midpoint, and k is the slope factor, where $V_h = +56.7$ mV and $k = 17.1$ mV, the mean values from the fits to data from each of five experiments. (B) Same for 30 nM VX-150m (mean $\pm$ S.D., $n = 5$). Solid line: $V_h = +3.7$ mV and $k = 13.1$ mV. (C) Same for 100 nM A-887826 (mean $\pm$ S.D., $n = 4$). Solid line: $V_h = +34.3$ mV and $k = 11.3$ mV.

changed when the voltage sensor moves from the “down” to the “up” position upon depolarization, offering a satisfying structural interpretation of the dramatic change in binding affinity produced by depolarization.

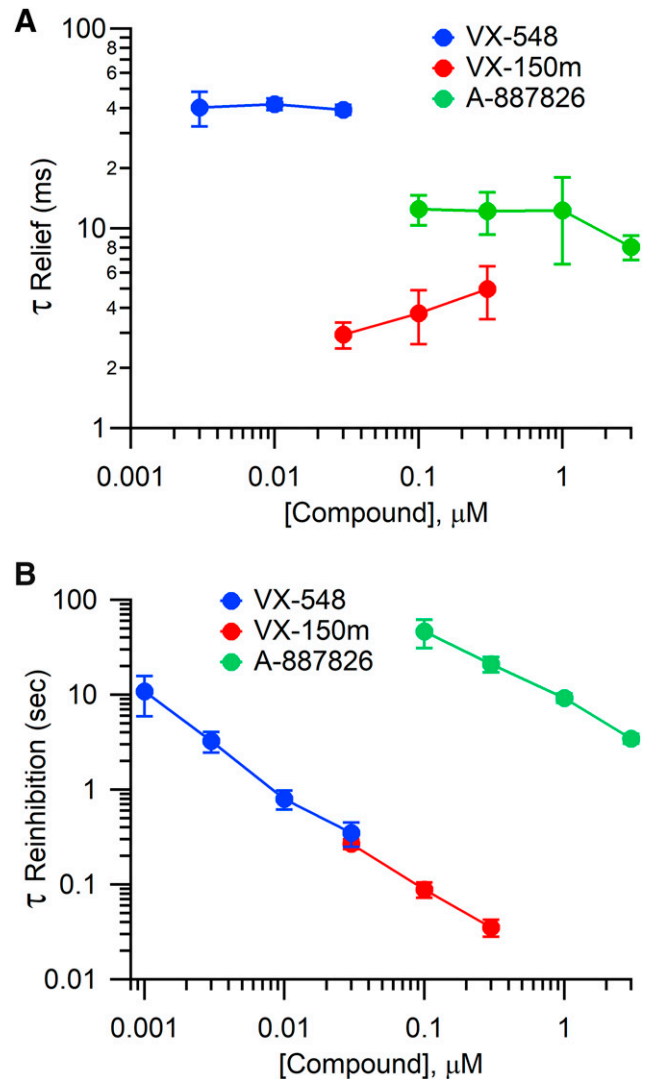


Fig. 9. Comparison of kinetics of voltage-dependent relief and reinhibition by VX-548, VX-150m, and A-887826. (A) Time constant of relief of inhibition at $+100$ mV measured as in Fig. 6 (Mean $\pm$ S.D.; $n = 7$ for 3 nM VX-548; $n = 8$ for 10 nM and 30 nM VX-548; $n = 6$ for VX-150m; $n = 3$ for 0.1 μ M A-887826, $n = 4$ for 0.3 μ M, 1 μ M, and 3 μ M A-887826). (B) Time constant of reinhibition at -100 mV after relief of inhibition at $+120$ mV, measured as in Fig. 7. (Mean $\pm$ S.D., $n = 8$ for VX-548, $n = 6$ for VX-150m, $n = 4$ for A-887826).

If the relief of depolarization occurs because of outward movement of the voltage sensor of domain II, why is the voltage-dependence of relief so different for VX-548, with a midpoint of relief of $+57$ mV, compared with VX-150m, with a midpoint of $+4$ mV? This could be explained by different relative affinities of compound binding to the resting and depolarized states of the channel. If compound binds more tightly to the resting state than to the depolarized state, it would stabilize the resting state relative to the depolarized state so that movement of the voltage sensor would be shifted to more depolarized voltages. This would be analogous to the action of drugs that bind with higher affinity to the inactivated state of a channel than to the resting state and would therefore shift the voltage-dependence of inactivation to more negative voltages. The voltage-dependence of relief of inhibition by drugs that stabilize the resting position of a voltage sensor

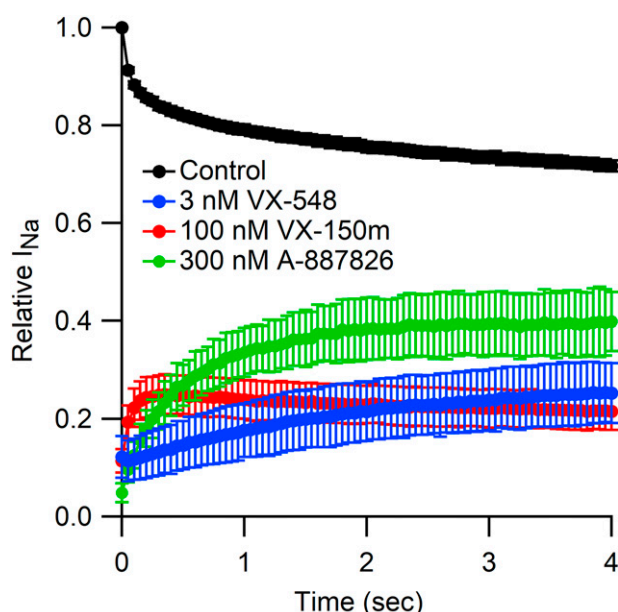


Fig. 10. Comparison of reverse use-dependence with VX-548, VX-150m, and A-887826. Use-dependent relief of inhibition by VX-548 (blue) compared with VX-150m (red, replotted from Fig. 2) and A-887826 (green). Mean $\pm$ S.D., $n = 5$ for VX-548, $n = 6$ for VX-150m, $n = 4$ for A-887826, $n = 14$ for control.

would be expected to be shifted in the depolarized direction compared with the voltage-dependence of movement of the sensor in the absence of drug, with the degree of shift greater with a greater difference in drug affinity for the resting versus depolarized state. The data therefore suggest that VX-548 has not only the highest affinity for the resting state of the channels, as reflected in the lowest IC_{50} for channel inhibition when applied at resting voltages, but also the most pronounced difference in binding to the resting versus depolarized states.

In principle, a drug that stabilizes the resting state of a channel relative to the activated state might shift the voltage-dependence of activation of the channel as measured by channel opening. This could be manifested as a shift in peak-conductance versus voltage relationship but only if drug-bound channels can actually open with sufficiently large depolarizations. This occurs with some gating-modifying toxins, where toxin-bound channels can open with large enough depolarizations, followed by slower dissociation of toxin from the toxin-bound open state of the channels (McDonough et al., 1997; Edgerton et al., 2008). In contrast, with both VX-150m and VX-548, the peak-conductance versus voltage relationship seemed to be simply scaled down (data not shown), as if drug-bound channels cannot open even with strong depolarizations. This appears to be an interesting difference between VX-548 (and VX-150m) and at least some toxins that act by stabilizing the resting state of channels, including ProTx-II (Edgerton et al., 2008).

Structures of mammalian voltage-dependent sodium channels by cryogenic electron microscopy (cryo-EM) presumably capture states of the channels that are at 0 mV for many minutes or hours, and so far there are no structures of mammalian sodium channels with voltage sensors in the resting

state. The data in Fig. 8A suggest that that the binding of VX-548 could stabilize the channel in a state at 0 mV with the domain II voltage sensors in the resting position since relief of inhibition is not evident until positive to 0 mV. However, a caveat is that the experimental determination of relief of inhibition using 200-millisecond depolarizations may not necessarily reflect the states of the channel occupied after hours at 0 mV. For VX-150m, voltage-dependent relief of inhibition is substantial at 0 mV, so one might predict that it would be more difficult to capture drug-bound states by cryo-EM with VX-150m than with VX-548.

So far, the only cryo-EM structure of a drug-bound Nav1.8 channel is with A-803467 (Huang et al., 2022). In the resolved structure, the A-803467 molecule is within the pore of the channel below the selectivity filter, with no evidence of binding to the domain II voltage sensor, which is in an “up” depolarized position in the resolved structure. Possibly A-803467 does also bind to the domain II voltage sensor at normal resting potentials but comes off the voltage sensor region when channels are at 0 mV. However, although A-803467 does show some reverse use-dependence (Browne et al., 2009a,b; Jo et al., 2023), it is far less dramatic than with A-887826, VX-150m, and VX-548, so the interactions of A-803467 with the channel may be different than these compounds.

Acknowledgments

The authors are grateful to Jinbo Lee for advice on the synthesis of VX-548 and to Robert Stewart for helpful discussion.

Data Availability

The authors declare that all of the data supporting the findings of this study are available within the paper and its Supplemental Material.

Authorship Contributions

Participated in research design: Vaelli, Fujita, Jo, Zhang, Osorno, Ma, Bean.

Conducted experiments: Vaelli, Fujita, Ma, Bean.

Contributed new reagents or analytic tools: Ma.

Performed data analysis: Vaelli, Bean.

Wrote or contributed to the writing of the manuscript: Vaelli, Ma, Bean.

References

- Bennett DL, Clark AJ, Huang J, Waxman SG, and Dib-Hajj SD (2019) The role of voltage-gated sodium channels in pain signaling. *Physiol Rev* **99**:1079–1151.
- Brown AD, Bagal SK, Blackwell P, Blakemore DC, Brown B, Bungay PJ, Corless M, Crawforth J, Fingas D, Fenwick DR, et al. (2019) The discovery and optimization of benzimidazoles as selective Nav1.8 blockers for the treatment of pain. *Bioorg Med Chem* **27**:230–239.
- Browne LE, Blaney FE, Yusaf SP, Clare JJ, and Wray D (2009a) Structural determinants of drugs acting on the Nav1.8 channel. *J Biol Chem* **284**:10523–10536.
- Browne LE, Clare JJ, and Wray D (2009b) Functional and pharmacological properties of human and rat Nav1.8 channels. *Neuropharmacology* **56**:905–914.
- Cardenas CA, Cardenas CG, de Armendi AJ, and Scroggs RS (2006) Carbamazepine interacts with a slow inactivation state of Nav1.8-like sodium channels. *Neurosci Lett* **408**:129–134.
- Edgerton GB, Blumenthal KM, and Hanck DA (2008) Evidence for multiple effects of ProTxII on activation gating in Nav1.5. *Toxicol* **52**:489–500.
- Gilchrist JM, Yang N-D, Jiang V, and Moyer BD (2024) Pharmacologic characterization of LTGO-33, a selective small molecule inhibitor of the voltage-gated sodium channel Nav1.8 with a unique mechanism of action. *Mol Pharmacol* **105**:233–249.
- Goodwin G and McMahon SB (2021) The physiological function of different voltage-gated sodium channels in pain. *Nat Rev Neurosci* **22**:263–274.
- Han C, Estacion M, Huang J, Vasylyev D, Zhao P, Dib-Hajj SD, and Waxman SG (2015) Human Nav1.8: enhanced persistent and ramp currents contribute to distinct firing properties of human DRG neurons. *J Neurophysiol* **113**:3172–3185.

- Han C, Huang J, and Waxman SG (2016) Sodium channel Nav1.8: Emerging links to human disease. *Neurology* **86**:473–483.
- Hijma HJ, Siebenga PS, de Kam ML, and Groeneveld GJ (2021) A phase 1, randomized, double-blind, placebo-controlled, crossover study to evaluate the pharmacodynamic effects of VX-150, a highly selective Nav1.8 inhibitor, in healthy male adults. *Pain Med* **22**:1814–1826.
- Hille B (1977) Local anesthetics: hydrophilic and hydrophobic pathways for the drug-receptor reaction. *J Gen Physiol* **69**:497–515.
- Houtmann S, Schombert B, Sanson C, Partiseti M, and Bohme GA (2017) Automated patch-clamp methods for the hERG cardiac potassium channel. *Methods Mol Biol* **1641**:187–199.
- Huang X, Jin X, Huang G, Huang J, Wu T, Li Z, Chen J, Kong F, Pan X, and Yan N (2022) Structural basis for high-voltage activation and subtype-specific inhibition of human Nav1.8. *Proc Natl Acad Sci U S A* **119**:e2208211119.
- Jarvis MF, Honore P, Shieh C-C, Chapman M, Joshi S, Zhang X-F, Kort M, Carroll W, Marron B, Atkinson R, et al. (2007) A-803467, a potent and selective Nav1.8 sodium channel blocker, attenuates neuropathic and inflammatory pain in the rat. *Proc Natl Acad Sci U S A* **104**:8520–8525.
- Jayakar S, Shim J, Jo S, Bean BP, Singeç I, and Woolf CJ (2021) Developing nociceptor-selective treatments for acute and chronic pain. *Sci Transl Med* **13**:eabj9837.
- Jo S, Zhang H-XB, and Bean BP (2023) Use-dependent relief of inhibition of Nav1.8 channels by A-887826. *Mol Pharmacol* **103**:221–229.
- Jones J, Correll DJ, Lechner SM, Jazic I, Miao X, Shaw D, Simard C, Osteen JD, Hare B, Beaton A, et al.; VX21-548-101 and VX21-548-102 Trial Groups (2023) Selective inhibition of Nav1.8 with VX-548 for acute pain. *N Engl J Med* **389**:393–405.
- Karkare R (2022) Solid dosage forms and dosing regimens comprising (2R,3S,4S,5R)-4-[[[3-(3,4-difluoro-2-methoxy-phenyl)-4,5-dimethyl-5-(trifluoromethyl) tetrahydrofuran-2-carbonyl]amino]pyridine-2-carboxamide. PCT WO 2022/256708 A1.
- Leffler A, Reckzeh J, and Nau C (2010) Block of sensory neuronal Na⁺ channels by the secretolytic ambroxol is associated with an interaction with local anesthetic binding sites. *Eur J Pharmacol* **630**:19–28.
- Marino M, Misuri L, and Broglioli D (2014) A new open source software for the calculation of the liquid junction potential between two solutions according to the stationary Nernst-Planck equation. *arXiv* **1403**:3640. [https://doi.org/10.48550/arXiv.1403.3640]
- Marx A, Backes C, Meese E, Lenhof H-P, and Keller A (2016) EDISON-WMW: exact dynamic programming solution of the Wilcoxon-Mann-Whitney test. *Genomics Proteomics Bioinformatics* **14**:55–61.
- McDonough SI, Mintz IM, and Bean BP (1997) Alteration of P-type calcium channel gating by the spider toxin omega-Aga-IVA. *Biophys J* **72**:2117–2128.
- Michel MC, Murphy TJ, and Motulsky HJ (2020) New author guidelines for displaying data and reporting data analysis and statistical methods in experimental biology. *Mol Pharmacol* **97**:49–60.
- Obeng S, Hiranita T, León F, McMahon LR, and McCurdy CR (2021) Novel approaches, drug candidates, and targets in pain drug discovery. *J Med Chem* **64**:6523–6548.
- Payne CE, Brown AR, Theile JW, Loucif AJC, Alexandrou AJ, Fuller MD, Mahoney JH, Antonio BM, Gerlach AC, Printzenhoff DM, et al. (2015) A novel selective and orally bioavailable Nav 1.8 channel blocker, PF-01247324, attenuates nociception and sensory neuron excitability. *Br J Pharmacol* **172**:2654–2670.
- Ryan MP (2020) Process for making pyridone amides and prodrugs thereof useful as modulators of sodium channels. PCT WO2020/206119 A1.
- Swartz KJ (2007) Tarantula toxins interacting with voltage sensors in potassium channels. *Toxicon* **49**:213–230.
- Swartz KJ and MacKinnon R (1997) Hanatoxin modifies the gating of a voltage-dependent K⁺ channel through multiple binding sites. *Neuron* **18**:665–673.
- Urru M, Muzzi M, Coppi E, Ranieri G, Buonvicino D, Camaioni E, Coppini R, Pugliese AM, Tanaka B, Estacion M, et al. (2020) Dextramipexole blocks Nav1.8 sodium channels and provides analgesia in multiple nociceptive and neuropathic pain models. *Pain* **161**:831–841.
- Yekkirala AS, Roberson DP, Bean BP, and Woolf CJ (2017) Breaking barriers to novel analgesic drug development. *Nat Rev Drug Discov* **16**:545–564.
- Zhang X-F, Shieh C-C, Chapman ML, Matulenko MA, Hakeem AH, Atkinson RN, Kort ME, Marron BE, Joshi S, Honore P, et al. (2010) A-887826 is a structurally novel, potent and voltage-dependent Na(v)1.8 sodium channel blocker that attenuates neuropathic tactile allodynia in rats. *Neuropharmacology* **59**:201–207.
- Zhou X, Dong X-W, Crona J, Maguire M, and Priestley T (2003) Vinpocetine is a potent blocker of rat Nav1.8 tetrodotoxin-resistant sodium channels. *J Pharmacol Exp Ther* **306**:498–504.

Address correspondence to: Dr. Bruce P. Bean, Department of Neurobiology, Harvard Medical School, 220 Longwood Avenue, Boston MA 02115. E-mail: bruce_bean@hms.harvard.edu
